Investigation 3.2 – Specific Heat of Pb.

RESULTS
- Suitable table or similar
- Units correct (3)
- Sig. fig correct.

CALCULATIONS
- Correct setting out
- Correct units & sig. fig.
- Correct calculation. (3)

DISCUSSION

Q4
- Comparison to accepted value. (2)
- % difference calculated.

Q5 Sources of greatest error
- Low ΔT value – has largest % uncertainty. (1)

Q6 Ways to reduce error
- Greater amount of Pb (2)
- Use Cu calorimeter in wood – better insulation.

Q7
- Dry metal – otherwise extra H_2O is added to the system. (1)
  – holds extra heat

Q8
- High specific heat ⇒ Stays hotter for longer – heat is lost slowly. (1)

Q9
- H_2O is better than metal. (2)
- Has a higher specific heat so stays hotter for longer.

CONCLUSION
- Statement of value of specific heat. (1)

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Invest. 4.2 – Heat of Fusion

RESULTS
- Suitable table or similar
- Correct units. (3)
- Correct sig. fig.

CALCULATIONS
- Correct setting out.
- Correct calculations.
- Correct units + sig. fig. (3)

DISCUSSION

Q3
- Sources of error (at least 2) – m_{ice}, m_{H_2O}, ΔT_{ice}, ΔT_{H_2O}.
- Estimate of % uncertainty for each. (3)
- Overall % uncertainty (should add all % uncertainties)

Q4. (a) • Impurities – lowers freezing point.

$m_w c_w \Delta T_w = m_i L_f + m_i c_w \Delta T_i$

(b) • Assume melting point is 0°C – ΔT will be larger (3)

(c) • Value would be lower.

Q5 • Above room temp ⇒ greater ΔT

⇒ negates the effect of heat entering from outside the system } EITHER (1)

Q6 • States accepted value.
• Calculates % difference. (2)

Q7 • Changes the E_p of the particles – allows them to move further apart and overcome some attractive forces. (1)

Q8 • Ice needs to melt first.
• Requires more heat, which is removed from the skin. (2)

CONCLUSION

• Statement of the latent heat of fusion. (1)

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